Supplementary Material for "TabSurv: Tabular Foundation Model for Breast Cancer Prognosis using Gene Expression Data"

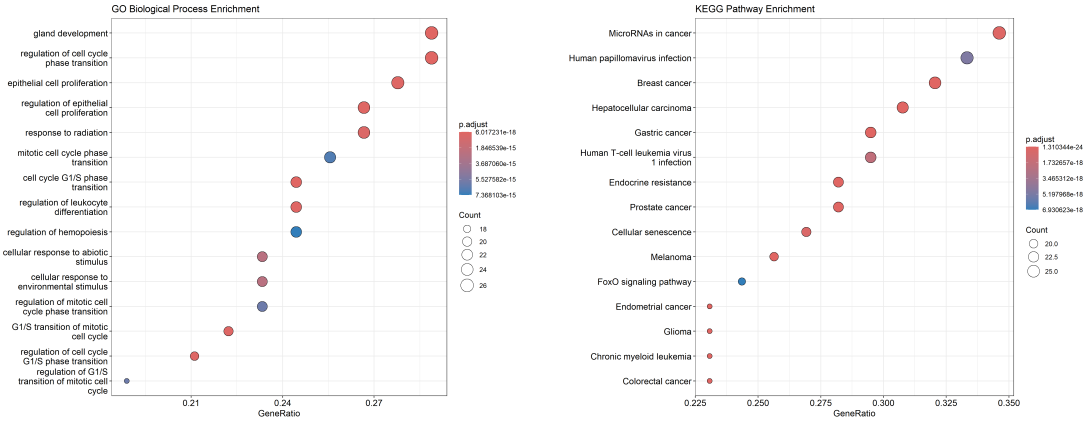
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1 Pathway Enrichment Analysis

To assess the biological coherence of the feature set used by TabSurv, we performed a pathway enrichment analysis on the 93 Nik-panel genes. Of these, 90 genes were successfully mapped to Entrez gene identifiers using the org.Hs.eg.db annotation database [1] and used for subsequent analysis. The enrichment of Gene Ontology (GO) [2] was carried out in terms of the biological process, and the enrichment of the KEGG [3] pathway was carried out, with adjusted p-values using the Benjamini-Hochberg method [4] (adjusted $p < 0.05$ considered significant).

1.1 Gene Ontology Enrichment

GO Biological Process analysis identified 1,453 significantly enriched terms (adjusted $p < 0.05$). The most enriched processes were gland development (26 genes; adjusted $p = 6.02 \times 10^{-18}$), regulation of the cell cycle phase transition (26 genes; adjusted $p = 7.39 \times 10^{-18}$) and the G1/S phase transition of the cell cycle (22 genes; adjusted $p = 8.46 \times 10^{-18}$). Additional terms highly enriched included epithelial cell proliferation (25 genes; adjusted $p = 1.75 \times 10^{-16}$), radiation response (24 genes; adjusted $p = 3.41 \times 10^{-16}$), and regulation of epithelial cell proliferation (24 genes; adjusted $p = 1.06 \times 10^{-16}$). Phosphatidylinositol 3-kinase signalling was significantly enriched (14 genes; adjusted $p = 1.09 \times 10^{-12}$), consistent with the inclusion of components of the PI3K pathway in the Nik-panel set. Cellular senescence (12 genes; adjusted $p = 1.71 \times 10^{-11}$) and negative regulation of the cell cycle (20 genes; adjusted $p = 6.44 \times 10^{-13}$) were also among the terms most enriched, reflecting the representation of tumour suppressor mechanisms. The top 15 enriched GO terms are shown in Figure S1a, and the complete list is available in SM-GO.



(a) **(b)**

Figure S1: Pathway enrichment analysis of the Nik-panel gene set. (a) Gene Ontology Biological Process enrichment. (b) KEGG pathway enrichment. Dot size indicates the number of genes; colour indicates adjusted p-value.

1.2 KEGG Pathway Enrichment

KEGG pathway analysis identified 68 significantly enriched pathways (adjusted $p < 0.05$). The most significantly enriched pathway was Breast cancer (29 genes; adjusted $p = 1.31 \times 10^{-24}$), followed by Endocrine resistance (22 genes; adjusted $p = 1.79 \times 10^{-24}$) and the PI3K-Akt signalling pathway (27 genes; adjusted $p = 3.63 \times 10^{-16}$). Additional cancer-related pathways with significant enrichment included Proteoglycans in cancer (21 genes; adjusted $p = 1.12 \times 10^{-14}$), Prostate cancer (16 genes; adjusted $p = 6.93 \times 10^{-14}$), and Hepatocellular carcinoma (17 genes; adjusted $p = 3.63 \times 10^{-13}$). Enrichment was also observed for Cellular senescence (16 genes; adjusted $p = 5.96 \times 10^{-12}$), p53 signalling pathway (11 genes; adjusted $p = 3.03 \times 10^{-10}$), and Cell cycle (13 genes; adjusted $p = 1.06 \times 10^{-9}$). Cross-cancer enrichment was evident, with significant results for Melanoma (12 genes; adjusted $p = 1.02 \times 10^{-11}$), Gastric cancer (14 genes; adjusted $p = 1.71 \times 10^{-10}$), and Colorectal cancer (12 genes; adjusted $p = 3.85 \times 10^{-10}$). The top 15 enriched KEGG pathways are shown in Figure S1b, and the complete list is available in SM-KEGG.

2 Breast Cancer Prognosis

2.1 In Distribution Experiment Results

Table S1: Benchmarking survival models under the in-distribution (InD) setting.

Dataset	TabSurv	DeepHS	DeepSurv	ENCox	LH	MTLR	PCHazard	PMF	RSF
METABRIC	**0.6265** ± 0.0079	0.5512 ± 0.0191	0.5923 ± 0.0076	0.5993 ± 0.0119	0.5872 ± 0.0218	0.5868 ± 0.0286	0.5929 ± 0.0148	0.5866 ± 0.0088	<u>0.6002</u> ± 0.0139
NKI	**0.6980** ± 0.0243	0.5588 ± 0.1030	0.6572 ± 0.0601	0.6290 ± 0.0236	0.6522 ± 0.0190	0.6391 ± 0.0341	<u>0.6584</u> ± 0.0176	0.6268 ± 0.0412	0.6574 ± 0.0469

Notes: DeepHS, DeepHitSingle; ENCox, ElasticNetCox; LH, Logistic Hazard; PMF, Probability Mass Function; MTLR, Multi-Task Logistic Regression; RSF, Random Survival Forest. Best-performing methods are shown in bold.

Table S2: In-distribution comparison (held-out METABRIC/NKI splits). ΔC: TabSurv_M minus baseline, averaged over five seeds, with 95% paired patient-level bootstrap CI.

Endpoint	Baseline	ΔC (95% CI)	p_{H}
RFS	DeepHS	+0.0753 (+0.0606, +0.0900)	0.0080†
RFS	DeepSurv	+0.0342 (+0.0243, +0.0446)	0.0080†
RFS	ENCox	+0.0272 (+0.0169, +0.0373)	0.0080†
RFS	LH	+0.0393 (+0.0289, +0.0500)	0.0080†
RFS	MTLR	+0.0397 (+0.0277, +0.0511)	0.0080†
RFS	PCHazard	+0.0336 (+0.0235, +0.0429)	0.0080†
RFS	PMF	+0.0399 (+0.0290, +0.0504)	0.0080†
RFS	RSF	+0.0263 (+0.0162, +0.0364)	0.0080†
DMFS	DeepHS	+0.1392 (+0.1019, +0.1753)	0.0080†
DMFS	DeepSurv	+0.0408 (+0.0114, +0.0693)	0.0120
DMFS	ENCox	+0.0690 (+0.0421, +0.0977)	0.0080†
DMFS	LH	+0.0458 (+0.0165, +0.0773)	0.0080†
DMFS	MTLR	+0.0589 (+0.0289, +0.0893)	0.0080†
DMFS	PCHazard	+0.0396 (+0.0123, +0.0695)	0.0160
DMFS	PMF	+0.0712 (+0.0413, +0.1063)	0.0080†
DMFS	RSF	+0.0406 (+0.0139, +0.0667)	0.0160

Notes: p_{H}: two-sided paired bootstrap p-value, $p = 2\min\{1 + \#(\Delta^* \leq 0),\ 1 + \#(\Delta^* \geq 0)\}/(B+1)$ with $B = 2{,}000$ resamples, Holm-adjusted across the 8 baselines within each endpoint. With $B = 2{,}000$ the smallest attainable unadjusted p is $2/(B+1) = 0.0010$ and the smallest attainable Holm-adjusted p is $8 \times 0.0010 = 0.0080$. Values marked † equal this bound, so the exact adjusted p-value may be smaller. All values are exact bootstrap p-values; no normal approximation is used.

For the in-distribution scenario, each dataset is partitioned using a single random split, with 50% of the samples allocated for training and 50% reserved for testing, using five random seeds (40–44) to ensure reproducibility. We evaluate on the 2 largest cohorts, one per endpoint: METABRIC for relapse-free survival (RFS), and NKI for distant metastasis-free survival (DMFS). Table S1 and S2 demonstrate that TabSurv achieves the highest C-index on both datasets (0.6265 on METABRIC and 0.6980 on NKI), outperforming all eight baseline survival models. The second-best method varies by dataset: RSF on METABRIC (0.6002) and PCHazard on NKI (0.6584), with TabSurv exceeding both by a margin of roughly 2–4 points in C-index.

Across both datasets, all baseline models maintain C-index values comfortably above 0.5, indicating no catastrophic failures on this benchmark; however, DeepHS shows the weakest and least stable performance, with a notably higher standard deviation on NKI (±0.1030) compared to all other methods, suggesting sensitivity to the specific train/test split and random seed.

2.2 Statistical Comparison Results

Table S3: Cohort-level ΔC (TabSurv minus baseline; mean over five seeds) with 95% paired patient-level bootstrap CIs (2,000 resamples).

Cohort	DeepHS	DeepSurv	ENCox	LH	MTLR	PCHazard	PMF	RSF
RFS								
GEO	+0.091 (+0.059, +0.122)	+0.034 (+0.015, +0.054)	+0.042 (+0.023, +0.061)	+0.054 (+0.034, +0.073)	+0.048 (+0.025, +0.069)	+0.037 (+0.020, +0.054)	+0.045 (+0.023, +0.064)	+0.028 (+0.004, +0.052)
GSE19783	+0.072 (-0.019, +0.161)	+0.008 (-0.044, +0.062)	+0.039 (-0.009, +0.090)	+0.037 (-0.028, +0.102)	+0.020 (-0.042, +0.081)	+0.024 (-0.032, +0.081)	+0.040 (-0.019, +0.099)	+0.034 (-0.038, +0.111)
GSE6532	+0.092 (+0.041, +0.141)	+0.011 (-0.022, +0.045)	+0.043 (+0.017, +0.069)	+0.037 (+0.008, +0.065)	+0.036 (+0.003, +0.069)	+0.043 (+0.017, +0.067)	+0.022 (-0.009, +0.054)	+0.040 (+0.003, +0.079)
TCGA500	+0.074 (-0.022, +0.167)	+0.019 (-0.045, +0.085)	-0.012 (-0.071, +0.050)	+0.027 (-0.032, +0.086)	+0.002 (-0.069, +0.077)	+0.007 (-0.043, +0.059)	+0.031 (-0.029, +0.091)	-0.013 (-0.079, +0.051)
UK	+0.136 (+0.074, +0.197)	+0.032 (-0.006, +0.070)	+0.077 (+0.042, +0.114)	+0.063 (+0.028, +0.099)	+0.052 (+0.012, +0.091)	+0.045 (+0.007, +0.081)	+0.062 (+0.025, +0.100)	+0.046 (-0.003, +0.095)
UPP	+0.085 (+0.015, +0.154)	+0.015 (-0.028, +0.059)	+0.040 (-0.000, +0.080)	+0.024 (-0.015, +0.064)	+0.012 (-0.034, +0.061)	+0.017 (-0.016, +0.051)	-0.002 (-0.042, +0.042)	-0.012 (-0.069, +0.045)
DMFS								
HEL	+0.109 (+0.044, +0.175)	+0.057 (+0.005, +0.118)	+0.056 (+0.006, +0.102)	+0.048 (+0.010, +0.092)	+0.073 (+0.025, +0.125)	+0.059 (+0.017, +0.105)	+0.046 (-0.003, +0.095)	+0.026 (-0.021, +0.070)
MAINZ	+0.087 (+0.041, +0.133)	+0.067 (+0.020, +0.113)	+0.063 (+0.017, +0.108)	+0.039 (+0.004, +0.072)	+0.046 (+0.016, +0.076)	+0.040 (+0.003, +0.075)	+0.062 (+0.029, +0.094)	+0.045 (+0.012, +0.076)
TRANSBIG	+0.039 (-0.005, +0.082)	-0.016 (-0.056, +0.028)	+0.037 (-0.002, +0.075)	+0.011 (-0.022, +0.045)	+0.012 (-0.022, +0.049)	+0.015 (-0.018, +0.049)	+0.049 (+0.012, +0.086)	+0.001 (-0.030, +0.032)
UNT	+0.090 (+0.016, +0.160)	+0.031 (-0.029, +0.090)	+0.051 (+0.005, +0.101)	+0.018 (-0.032, +0.073)	+0.032 (-0.013, +0.083)	+0.020 (-0.027, +0.066)	+0.063 (+0.008, +0.119)	+0.035 (-0.011, +0.083)

Table S3 gives the paired difference in C-index, $\Delta C = C(\text{TabSurv}) - C(\text{baseline})$, averaged over the five random seeds, together with a 95% confidence interval (CI) from a paired patient-level bootstrap. In each of the 2,000 bootstrap resamples, the same patients were scored by both models across all seeds, so the intervals account for the correlation between the two models' predictions on the same patients. A positive ΔC means that TabSurv ranked patients more accurately than the baseline. These cohort-level intervals are descriptive and are not adjusted for multiplicity. Formal inference is based on the random-effects meta-analysis with Holm adjustment described in the main text.

2.3 Accuracy and Calibration of Predicted Survival

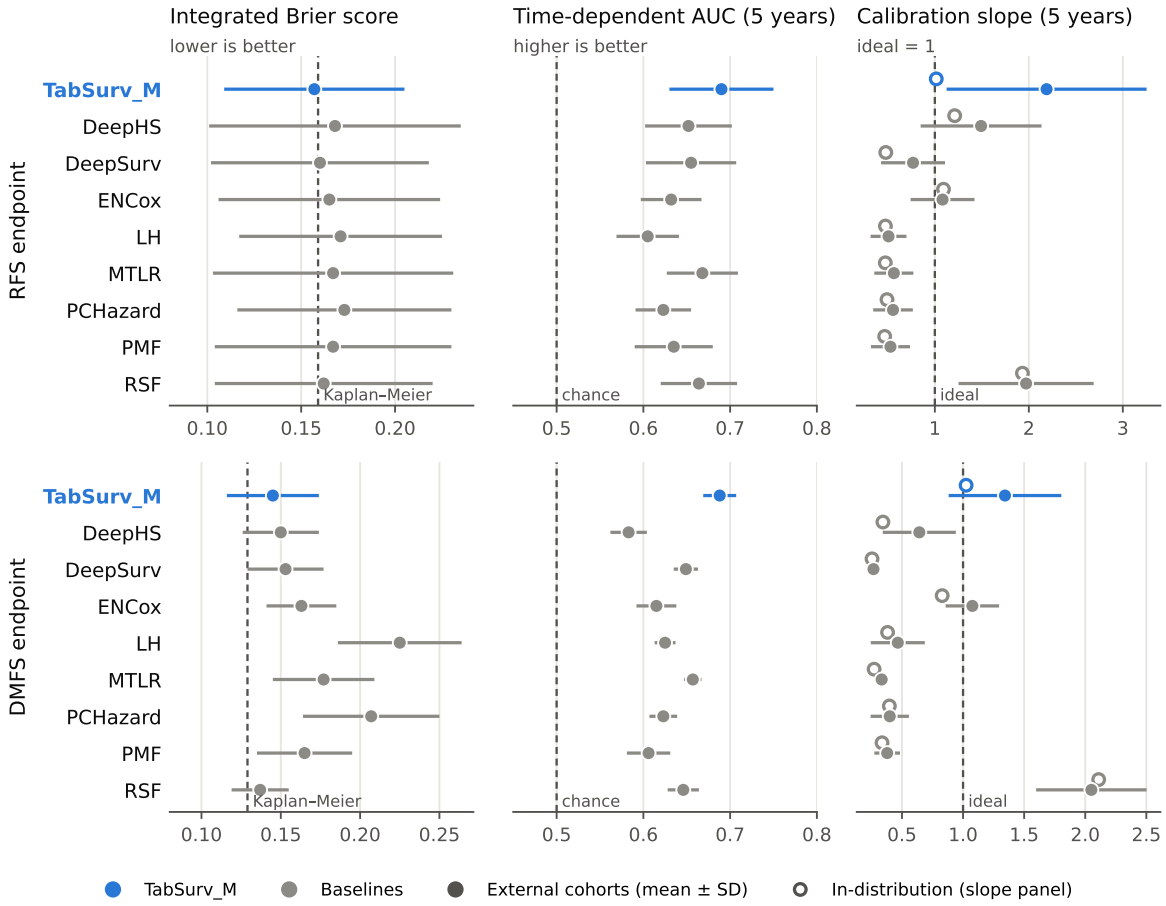


Figure S2: Accuracy and calibration of predicted survival for TabSurv and the eight baselines. Left: integrated Brier score (dashed line: covariate-free Kaplan–Meier reference). Middle: time-dependent AUC at 5 years (dashed line: chance). Right: calibration slope at 5 years (dashed line: ideal value of 1). Filled circles and bars show the mean ± SD across the external cohorts (RFS, top; DMFS, bottom); open circles show the in-distribution calibration slope.

We additionally assessed the accuracy and calibration of each model's predicted survival function $S(t \mid x)$. For TabSurv, $S(t \mid x) = 1 - F(t \mid x)$, where $F(t \mid x)$ is obtained from the TabPFN predictive distribution (99 predictive quantiles) and averaged over the M Stage-2 models, i.e. the pooled predictive distribution across imputations. The baselines use their own survival functions. We report: (i) the integrated Brier score (IBS) [5], with inverse-probability-of-censoring weights estimated in each evaluation cohort, integrated from the 5th percentile of follow-up to the earlier of 10 years and the 80th percentile of follow-up, together with a covariate-free Kaplan–Meier reference; (ii) prediction-error curves (the Brier score over time); (iii) the cumulative/dynamic time-dependent AUC [6] at 3, 5 and 10 years; (iv) calibration curves at 5 years, comparing mean predicted risk with Kaplan–Meier observed risk within quintiles of predicted risk; and (v) the calibration slope, estimated by Cox regression of the outcome on $\log\{-\log S(t_h \mid x)\}$ with follow-up truncated at horizon t_h [7], together with the observed-to-expected (O/E) ratio. A calibration slope of 1 indicates ideal calibration; values below 1 indicate over-dispersed predictions (risk differences between patients are exaggerated) and values above 1 indicate under-dispersed predictions. A horizon was not evaluated in a cohort whose follow-up did not extend beyond it or with fewer than 10 events before it. Predicted event times are interpreted as risk rankings, not as validated estimates of individual event times.

Table S4: Accuracy and calibration of predicted survival on the external cohorts (RFS: 6 cohorts, DMFS: 4 cohorts). IBS: integrated Brier score (lower is better), integrated up to 10 years or the 80th percentile of follow-up; the Kaplan–Meier row is a covariate-free reference. AUC(5y): cumulative/dynamic time-dependent AUC at 5 years. Slope(5y): calibration slope at 5 years (ideal 1; < 1 indicates over-dispersed, > 1 under-dispersed predictions). Values are mean ± SD across cohorts after averaging over five seeds; [k] marks metrics available in only k cohorts because follow-up did not extend beyond 5 years.

	RFS endpoint			DMFS endpoint		
Model	IBS	AUC(5y)	Slope(5y)	IBS	AUC(5y)	Slope(5y)
TabSurv_M	0.157 ± 0.048	0.690 ± 0.060	2.19 ± 1.06	0.145 ± 0.029	0.688 ± 0.019 [3]	1.34 ± 0.46 [3]
DeepHS	0.168 ± 0.067	0.652 ± 0.050	1.49 ± 0.64	0.150 ± 0.024	0.583 ± 0.021 [3]	0.64 ± 0.30 [3]
DeepSurv	0.160 ± 0.058	0.655 ± 0.052	0.77 ± 0.34	0.153 ± 0.024	0.649 ± 0.014 [3]	0.27 ± 0.02 [3]
ENCox	0.165 ± 0.059	0.632 ± 0.035	1.08 ± 0.34	0.163 ± 0.022	0.615 ± 0.023 [3]	1.08 ± 0.22 [3]
LH	0.171 ± 0.054	0.605 ± 0.036	0.51 ± 0.19	0.225 ± 0.039	0.625 ± 0.012 [3]	0.47 ± 0.22 [3]
MTLR	0.167 ± 0.064	0.668 ± 0.041	0.56 ± 0.21	0.177 ± 0.032	0.657 ± 0.010 [3]	0.33 ± 0.03 [3]
PCHazard	0.173 ± 0.057	0.623 ± 0.032	0.55 ± 0.21	0.207 ± 0.043	0.623 ± 0.016 [3]	0.40 ± 0.16 [3]
PMF	0.167 ± 0.063	0.635 ± 0.045	0.53 ± 0.21	0.165 ± 0.030	0.606 ± 0.025 [3]	0.38 ± 0.10 [3]
RSF	0.162 ± 0.058	0.664 ± 0.044	1.97 ± 0.72	0.137 ± 0.018	0.646 ± 0.018 [3]	2.05 ± 0.45 [3]
Kaplan–Meier (reference)	0.159 ± 0.051	0.500	–	0.129 ± 0.013	0.500	–

Figure S2 and numerical values in Table S4 compare TabSurv with the eight baselines on three measures of accuracy and calibration. The left column shows the integrated Brier score (lower is better), integrated up to 10 years or the 80th percentile of follow-up, whichever is earlier; the dashed line marks the covariate-free Kaplan–Meier reference. The middle column shows the cumulative/dynamic time-dependent AUC at 5 years (higher is better); the dashed line marks chance (0.5). The right column shows the calibration slope at 5 years; the dashed line marks the ideal value of 1, and values above or below 1 indicate under- or over-dispersed predictions, respectively. Filled markers and whiskers show the mean ± SD across external cohorts after averaging over five seeds; open markers in the right column show the in-distribution slope on the held-out METABRIC and NKI splits. The 5-year DMFS metrics are based on three cohorts because follow-up in HEL is only five years.

Numerical values for all horizons and for the in-distribution setting are given in Supplementary Table S5. Calibration and prediction-error curves for every cohort are shown in Supplementary Figures S3–S6. Figures S3 and S5 compare the predicted 5-year risk with the Kaplan–Meier observed risk in risk groups for the RFS and DMFS endpoints, with the in-distribution cohort shown first and each external cohort in its own panel. Figures S4 and S6 show the Brier score over time for every cohort against the covariate-free Kaplan–Meier reference.

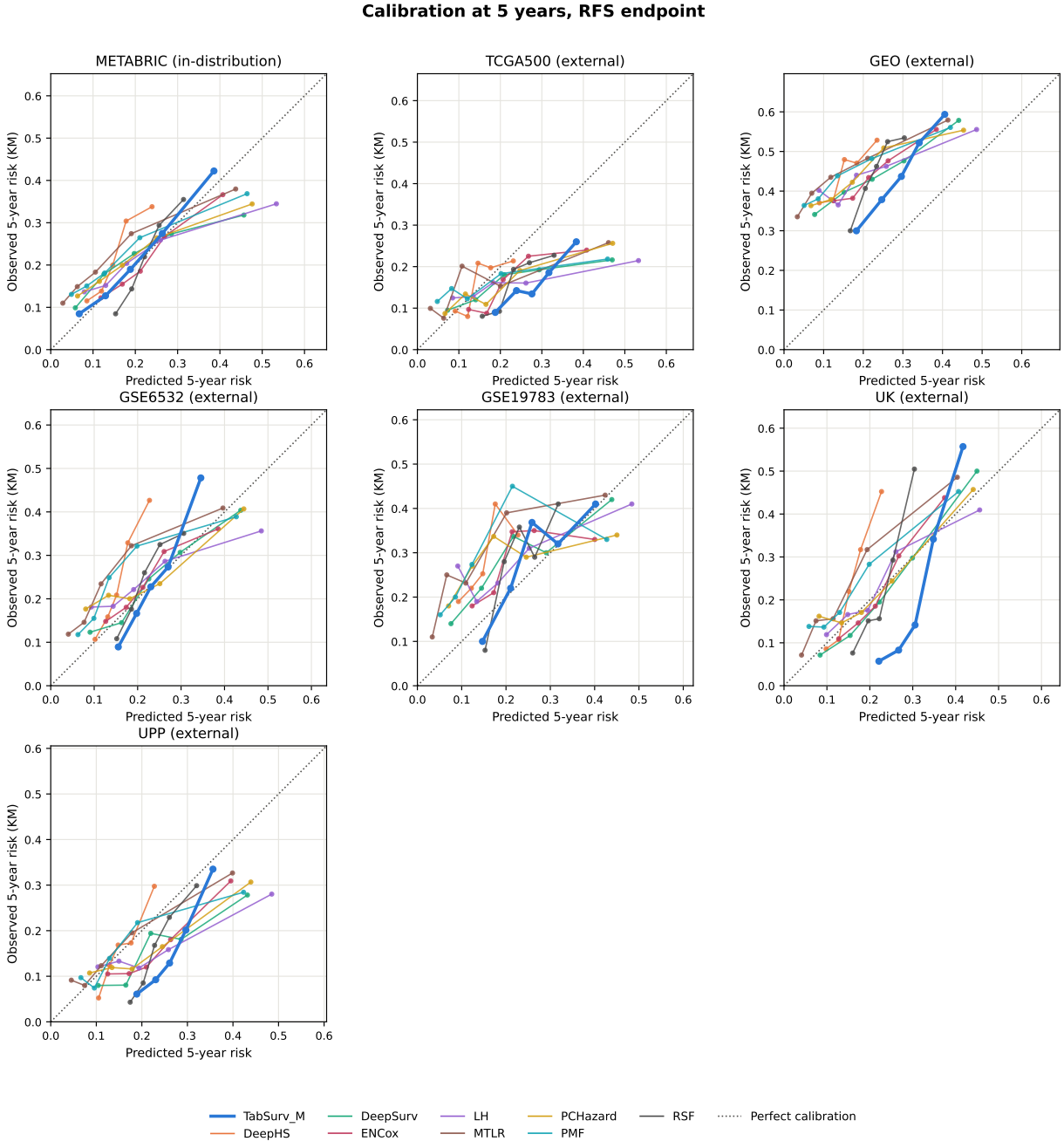


Figure S3: Calibration of predicted 5-year RFS risk in every cohort. Each point is a risk group: mean predicted risk of an event by 5 years against the Kaplan–Meier observed risk in that group, averaged over five seeds. The dotted diagonal marks perfect calibration. TabSurv is shown as a thick blue line. Panels marked as having too short follow-up have no patients at risk beyond 5 years.

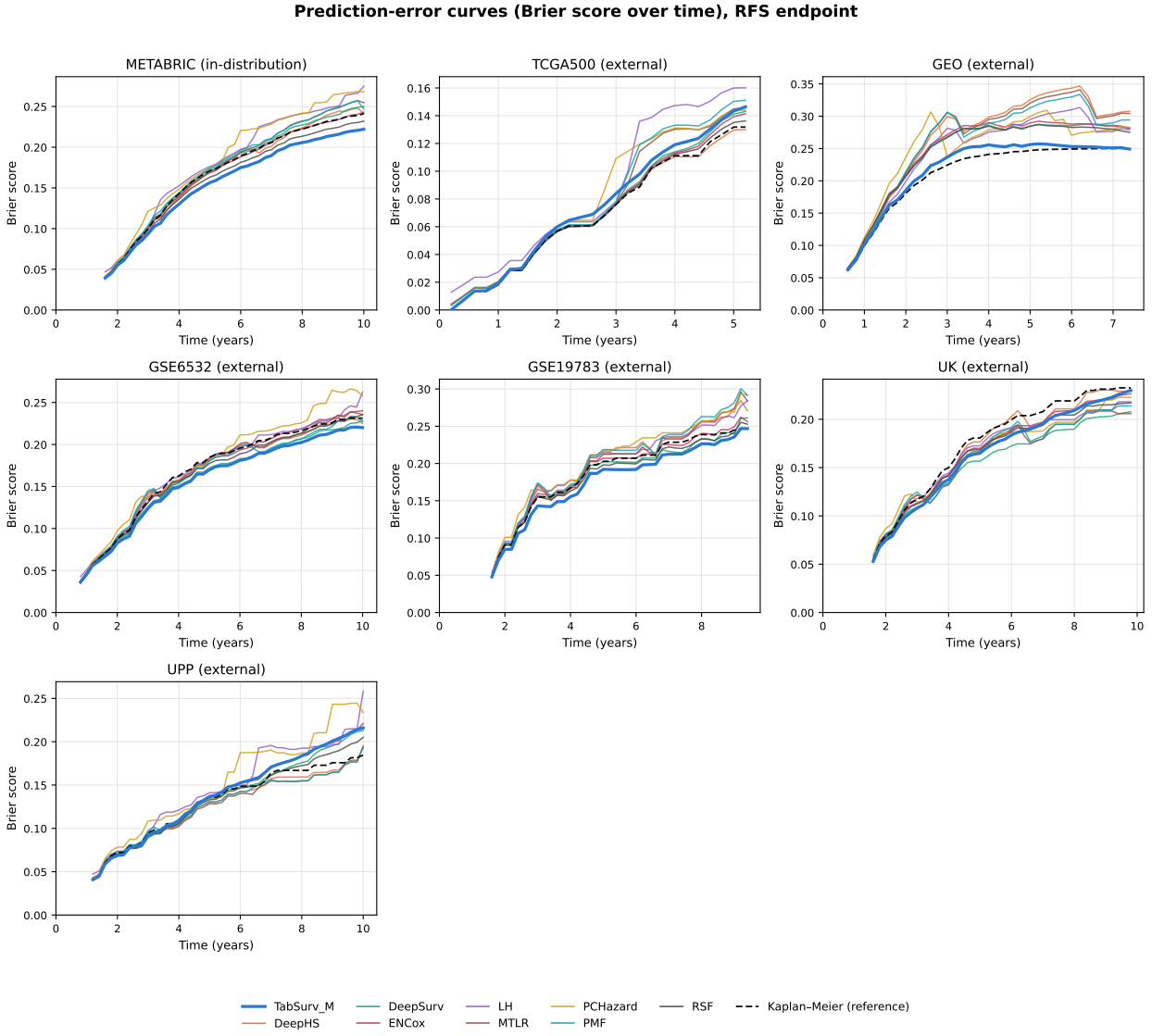


Figure S4: Prediction-error curves for the RFS endpoint in every cohort: Brier score over time (lower is better), averaged over five seeds. The dashed black line is the covariate-free Kaplan–Meier reference; a model curve below it indicates that the covariates improve prediction at that time. TabSurv is shown as a thick blue line.

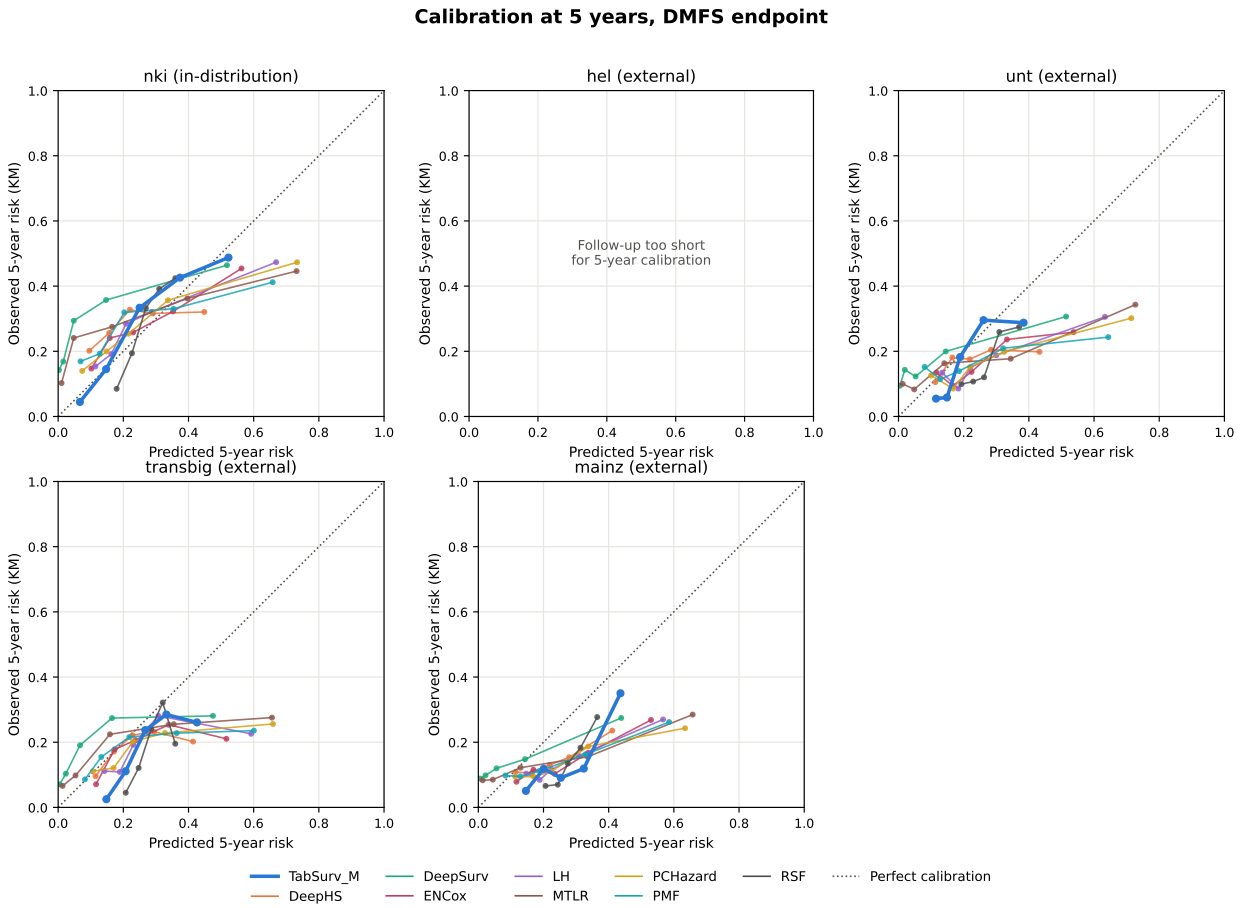


Figure S5: Calibration of predicted 5-year DMFS risk in every cohort. Each point is a risk group: mean predicted risk of an event by 5 years against the Kaplan–Meier observed risk in that group, averaged over five seeds. The dotted diagonal marks perfect calibration. TabSurv is shown as a thick blue line. Panels marked as having too short follow-up have no patients at risk beyond 5 years. Cohort names are shown in lowercase to mark the DMFS endpoint.

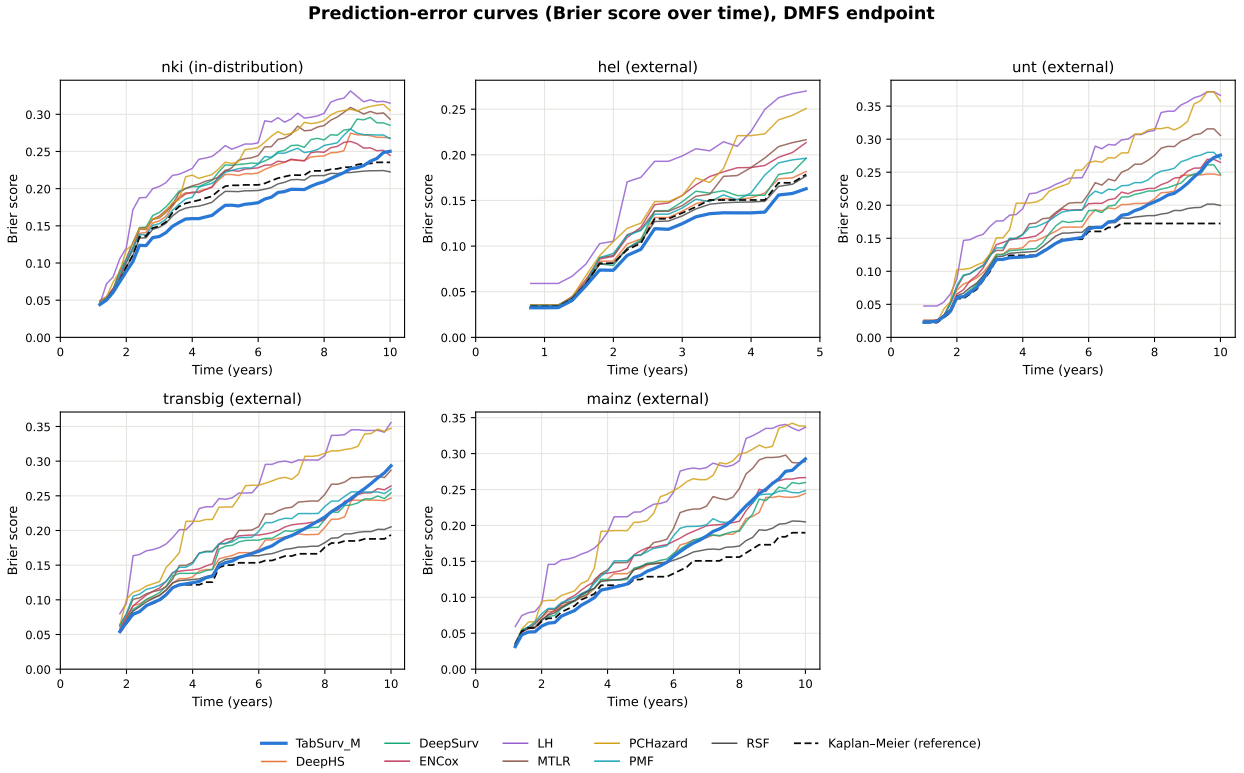


Figure S6: Prediction-error curves for the DMFS endpoint in every cohort: Brier score over time (lower is better), averaged over five seeds. The dashed black line is the covariate-free Kaplan–Meier reference; a model curve below it indicates that the covariates improve prediction at that time. TabSurv is shown as a thick blue line. Cohort names are shown in lowercase to mark the DMFS endpoint.

Time-dependent discrimination. TabSurv achieved the highest time-dependent AUC on both endpoints, at 3 years (RFS 0.714 vs. 0.491–0.676 for the baselines; DMFS 0.712 vs. 0.578–0.679) and at 5 years (RFS 0.690 vs. 0.605–0.668; DMFS 0.688 vs. 0.583–0.657; Figure S2, middle column), consistent with the C-index results.

Overall accuracy. On the RFS cohorts, TabSurv had the lowest IBS (0.157; baselines 0.160–0.173) and was the only model below the Kaplan–Meier reference (0.159), although by a small margin. On the DMFS cohorts, TabSurv had the second-lowest IBS (0.145, after RSF 0.137), but no model improved on the Kaplan–Meier reference (0.129; Figure S2, left column). This reflects systematic over-estimation of absolute risk in the external DMFS cohorts, whose event rates (21–31%) are lower than in the NKI training cohort (34%); the 5-year observed-to-expected ratio of TabSurv was 0.67.

Calibration. In distribution, TabSurv was well calibrated, with 5-year calibration slopes of 1.02 (RFS) and 1.03 (DMFS), the closest to the ideal value of all models (Figure S2, open markers), and the lowest IBS, below the Kaplan–Meier reference (RFS 0.159 vs. 0.173; DMFS 0.173 vs. 0.188). On the external cohorts, its calibration slope exceeded 1 (RFS 2.19 ± 1.06; DMFS 1.34 ± 0.46; filled markers): the ranking of patients transferred across cohorts, but the spread of predicted absolute risks was too narrow. The baselines also departed from ideal calibration, and the direction differed by model. DeepSurv was over-dispersed on both endpoints (slopes 0.77 and 0.27), meaning its predictions were too extreme. RSF was under-dispersed (slopes 1.97 and 2.05), as was DeepHS on RFS (slope 1.49), meaning their predictions were too narrow. ENCox was closest to ideal, with a slope of 1.08 on both endpoints. Hence, under distribution shift, TabSurv's predicted survival should be used for risk stratification, and absolute risk estimates should be recalibrated in the target population (e.g. by re-estimating the calibration intercept and slope) before being interpreted as individual survival probabilities.

Table S5: Accuracy and calibration of predicted survival, all horizons. IBS: integrated Brier score (lower is better), with the covariate-free Kaplan–Meier (KM) reference in each block heading. AUC(t): cumulative/dynamic time-dependent AUC at t years. Slope and O/E: calibration slope (ideal 1) and observed-to-expected ratio (ideal 1) at 5 years. In distribution: mean ± SD over five seeds on the held-out METABRIC/NKI splits. External: mean ± SD across cohorts after averaging over five seeds; [k] marks metrics available in only k cohorts because follow-up did not extend beyond the horizon.

Model	IBS	AUC(3y)	AUC(5y)	AUC(10y)	Slope(5y)	O/E(5y)
RFS, in distribution (KM IBS 0.173 ± 0.004)						
TabSurv	0.159 ± 0.003	0.718 ± 0.022	0.701 ± 0.009	0.668 ± 0.011	1.015 ± 0.081	1.060 ± 0.077
DeepHS	0.174 ± 0.005	0.546 ± 0.061	0.650 ± 0.014	0.582 ± 0.024	1.211 ± 0.336	1.430 ± 0.134
DeepSurv	0.176 ± 0.003	0.609 ± 0.023	0.630 ± 0.020	0.612 ± 0.018	0.478 ± 0.059	0.978 ± 0.084
ENCox	0.172 ± 0.011	0.641 ± 0.029	0.648 ± 0.027	0.616 ± 0.016	1.092 ± 0.664	0.934 ± 0.075
LH	0.186 ± 0.004	0.542 ± 0.016	0.629 ± 0.014	0.617 ± 0.034	0.474 ± 0.047	0.938 ± 0.137
MTLR	0.177 ± 0.006	0.592 ± 0.063	0.663 ± 0.019	0.625 ± 0.030	0.473 ± 0.068	1.350 ± 0.216
PCHazard	0.190 ± 0.006	0.637 ± 0.019	0.633 ± 0.011	0.616 ± 0.020	0.491 ± 0.084	1.076 ± 0.322
PMF	0.179 ± 0.004	0.515 ± 0.075	0.642 ± 0.023	0.613 ± 0.023	0.466 ± 0.091	1.186 ± 0.176
RSF	0.166 ± 0.003	0.697 ± 0.031	0.670 ± 0.012	0.622 ± 0.026	1.932 ± 0.395	0.965 ± 0.070
RFS, external cohorts (KM IBS 0.159 ± 0.051)						
TabSurv	0.157 ± 0.048	0.714 ± 0.065	0.690 ± 0.060	0.620 ± 0.071 [5]	2.189 ± 1.064	0.921 ± 0.356
DeepHS	0.168 ± 0.067	0.578 ± 0.030	0.652 ± 0.050	0.565 ± 0.057 [5]	1.492 ± 0.643	1.637 ± 0.672
DeepSurv	0.160 ± 0.058	0.658 ± 0.041	0.655 ± 0.052	0.597 ± 0.070 [5]	0.768 ± 0.340	1.070 ± 0.446
ENCox	0.165 ± 0.059	0.654 ± 0.033	0.632 ± 0.035	0.580 ± 0.047 [5]	1.082 ± 0.340	1.095 ± 0.457
LH	0.171 ± 0.054	0.518 ± 0.019	0.605 ± 0.036	0.591 ± 0.072 [5]	0.507 ± 0.191	1.117 ± 0.478
MTLR	0.167 ± 0.064	0.620 ± 0.032	0.668 ± 0.041	0.620 ± 0.065 [5]	0.564 ± 0.208	1.566 ± 0.634
PCHazard	0.173 ± 0.057	0.639 ± 0.042	0.623 ± 0.032	0.608 ± 0.046 [5]	0.555 ± 0.211	1.228 ± 0.517
PMF	0.167 ± 0.063	0.491 ± 0.042	0.635 ± 0.045	0.608 ± 0.075 [5]	0.528 ± 0.208	1.409 ± 0.579
RSF	0.162 ± 0.058	0.676 ± 0.037	0.664 ± 0.044	0.593 ± 0.067 [5]	1.971 ± 0.719	1.101 ± 0.453
DMFS, in distribution (KM IBS 0.188 ± 0.005)						
TabSurv	0.173 ± 0.007	0.757 ± 0.034	0.741 ± 0.033	0.725 ± 0.046	1.025 ± 0.238	1.051 ± 0.114
DeepHS	0.205 ± 0.031	0.589 ± 0.157	0.566 ± 0.123	0.569 ± 0.122	0.344 ± 0.318	1.286 ± 0.506
DeepSurv	0.218 ± 0.019	0.714 ± 0.050	0.669 ± 0.084	0.660 ± 0.095	0.255 ± 0.130	2.433 ± 1.493
ENCox	0.206 ± 0.027	0.679 ± 0.039	0.645 ± 0.026	0.639 ± 0.057	0.829 ± 0.708	1.223 ± 0.727
LH	0.251 ± 0.090	0.650 ± 0.047	0.646 ± 0.039	0.669 ± 0.058	0.382 ± 0.209	1.424 ± 0.759
MTLR	0.226 ± 0.017	0.680 ± 0.017	0.667 ± 0.049	0.649 ± 0.084	0.272 ± 0.047	1.093 ± 0.257
PCHazard	0.232 ± 0.041	0.666 ± 0.032	0.666 ± 0.016	0.672 ± 0.055	0.397 ± 0.111	1.164 ± 0.506
PMF	0.210 ± 0.011	0.664 ± 0.094	0.627 ± 0.069	0.621 ± 0.069	0.337 ± 0.216	1.037 ± 0.229
RSF	0.182 ± 0.009	0.725 ± 0.041	0.679 ± 0.074	0.655 ± 0.074	2.110 ± 0.745	1.146 ± 0.426
DMFS, external cohorts (KM IBS 0.129 ± 0.013)						
TabSurv	0.145 ± 0.029	0.712 ± 0.085	0.688 ± 0.019 [3]	0.618 ± 0.040 [3]	1.343 ± 0.461 [3]	0.674 ± 0.139 [3]
DeepHS	0.150 ± 0.024	0.578 ± 0.032	0.583 ± 0.021 [3]	0.548 ± 0.015 [3]	0.642 ± 0.299 [3]	0.730 ± 0.080 [3]
DeepSurv	0.153 ± 0.024	0.655 ± 0.041	0.649 ± 0.014 [3]	0.598 ± 0.016 [3]	0.267 ± 0.020 [3]	1.806 ± 0.097 [3]
ENCox	0.163 ± 0.022	0.636 ± 0.054	0.615 ± 0.023 [3]	0.569 ± 0.015 [3]	1.076 ± 0.219 [3]	0.743 ± 0.085 [3]
LH	0.225 ± 0.039	0.624 ± 0.037	0.625 ± 0.012 [3]	0.589 ± 0.021 [3]	0.466 ± 0.221 [3]	0.986 ± 0.100 [3]
MTLR	0.177 ± 0.032	0.661 ± 0.050	0.657 ± 0.010 [3]	0.599 ± 0.009 [3]	0.334 ± 0.029 [3]	0.687 ± 0.064 [3]
PCHazard	0.207 ± 0.043	0.600 ± 0.031	0.623 ± 0.016 [3]	0.593 ± 0.015 [3]	0.400 ± 0.157 [3]	0.783 ± 0.073 [3]
PMF	0.165 ± 0.030	0.611 ± 0.043	0.606 ± 0.025 [3]	0.563 ± 0.026 [3]	0.379 ± 0.105 [3]	0.632 ± 0.056 [3]
RSF	0.137 ± 0.018	0.679 ± 0.059	0.646 ± 0.018 [3]	0.588 ± 0.019 [3]	2.050 ± 0.452 [3]	0.644 ± 0.075 [3]

3 Treatment Recommendation Evaluation

3.1 Comparison of TabSurv Recommendation performance across Nik-panel with two comparator panels

The two comparator panels were also used exactly as published:

1. **Mills-panel**, based on clinically actionable and biologically relevant genes reported by Mills et al. [8], focusing on genes associated with targeted therapy response and breast cancer progression.
2. **Pereira-panel**, derived from the comprehensive integrative genomic analysis of breast cancer by Pereira et al. [9], which identified recurrently mutated driver genes across a large breast cancer cohort.

Table S6: Comparison of Delta RMST (years) between the *Followed* and *Not Followed* groups across the three gene panels under the TabSurv recommendation framework.

Gene Set	**Δ RMST (mean $\pm$ std)**
Nik-panel	1.261 ± 1.992
Mills-panel	0.469 ± 0.576
Pereira-panel	0.693 ± 0.246

Δ RMST denotes the difference between *Followed* Recommendation and *Not Followed* Recommendation groups.

Table S6 summarises the difference in mean survival time between patients whose treatment did and did not follow the TabSurv recommendation, for each of the three gene sets. The mean difference between the Followed and Not Followed groups was 1.261 ± 1.992 years for the *Nik-panel*, 0.693 ± 0.246 years for the *Pereira-panel* and 0.469 ± 0.576 years for the *Mills-panel*. Given the large variability across splits, particularly for the *Nik-panel*, these differences cannot be distinguished from one another, and the comparison does not indicate that any one panel provides a stronger signal than the others. All three panels were fixed before analysis and are reported here as a descriptive comparison; none was selected on the basis of these results. We note that the *Pereira-panel* was itself derived from METABRIC: Pereira et al. identified its genes from targeted sequencing of the METABRIC tumours. Only their mutation data, not the expression values, survival outcomes or treatment information used in our analysis, contributed to its definition, and we use it solely as a published comparator.

3.2 Genes of the Nik-panel Used in the Experiments

Table S7 lists the 93 driver genes of the Nik-panel and highlights the 72 that are available in the METABRIC expression data after gene matching. Genes were matched by the symbols used in the source publication; genes listed under former symbols were not matched.

Table S7: The 93 driver genes of the Nik-panel, listed in order of decreasing mutation frequency.

TP53	***PIK3CA***	***MYC***	***CCND1***	***PTEN***	***ERBB2***	*ZNF703*	***FGFR1***
GATA3	***RB1***	*MAP3K1*	***MAP2K4***	***ZNF217***	***CDH1***	*MLL3*	*ARID1B*
CDKN2A	*MLLT4*	***AKT1***	***FBXW7***	***ARID1A***	***CCND3***	***CBFB***	***MDM2***
CCNE1	***CDKN2B***	***NCOR1***	***SF3B1***	***SPEN***	***TBX3***	*IGF1R*	***BRCA2***
NF1	***PIK3R1***	***EGFR***	*KRAS*	***ESR1***	***FOXA1***	***MED23***	***CDK6***
NOTCH2	***AKT2***	***BRCA1***	***CTCF***	***KDM6A***	***SETD2***	***CREBBP***	***DNMT3A***
FOXP1	*MLL2*	*RUNX1*	***USP9X***	***XBP1***	***PDGFRA***	***ATR***	***ERBB3***
FGFR2	*PALB2*	***RHOA***	***SMAD4***	***ATM***	*ATRX*	***AXIN1***	***BCOR***
CDKN1B	***CUX1***	***GNAS***	***MLH1***	***NOTCH1***	*PHF6*	***SMARCA4***	***STAG2***
ZFP36L1	***APC***	***CASP8***	*CBLB*	***CNOT3***	*ECT2L*	***MEN1***	***MSH2***
NRAS	*PBRM1*	*PMS2*	***STK11***	*TET2*	***ASXL1***	*BRAF*	*BUB1B*
CIC	***ERCC4***	***HRAS***	***NEF2***	*PRDM1*	*PREX2*		

Notes: The chromosome 8 amplicon (*ZNF703*/*FGFR1*), which is a single driver in the original list, is shown here as two separate genes, giving 94 genes in total. 72 genes in **bold** were included in the experiment.

3.3 Kaplan-Meier Curve of TabSurv Recommendation with Baseline Methods

Figure S7 and Figure S8 present Kaplan–Meier survival curves comparing patients whose received treatment matched the recommendation (Followed) with those whose treatment did not (Not Followed), using the curated set of 72 biologically important genes and all genes, respectively. The curves pool the test sets of the five random splits and are shown as descriptive summaries. We do not report log-rank p-values for these curves, for two reasons. First, the log-rank test only assesses whether the two survival curves differ; it does not indicate which curve lies above the other, whereas the quantity of interest here is the direction of the difference, namely whether the Followed curve lies above the Not Followed curve. Second, because the 70/30 test sets overlap across splits, some patients contribute to more than one split, so a test on the pooled curves would treat repeated observations as independent and overstate significance. The direction and size of the separation are instead quantified by the difference in restricted mean survival time (RMST) between the Followed and Not Followed groups: a positive difference means that the Followed curve lies above the Not Followed curve on average, i.e. that patients whose treatment matched the recommendation survived longer up to the prespecified horizon.

With the Nik-panel, the Followed group of TabSurv has higher survival than the Not Followed group throughout the 20-year follow-up, with the clearest separation in the first ten years. With all genes, the two TabSurv curves nearly overlap, with at most a small, mostly early, advantage for the Followed group. The baselines show no consistent long-term pattern: for DeepSurv, RSF and ENCox (all genes), the Not Followed group survives as well as or better than the Followed group, while LH, PMF, PCHazard, MTLR (all genes) and BITES show an early advantage for the Followed group that narrows or reverses by 10–17 years. Overall, TabSurv shows the most consistent separation in favour of the Followed group with the Nik-panel, but the difference is modest with all genes.

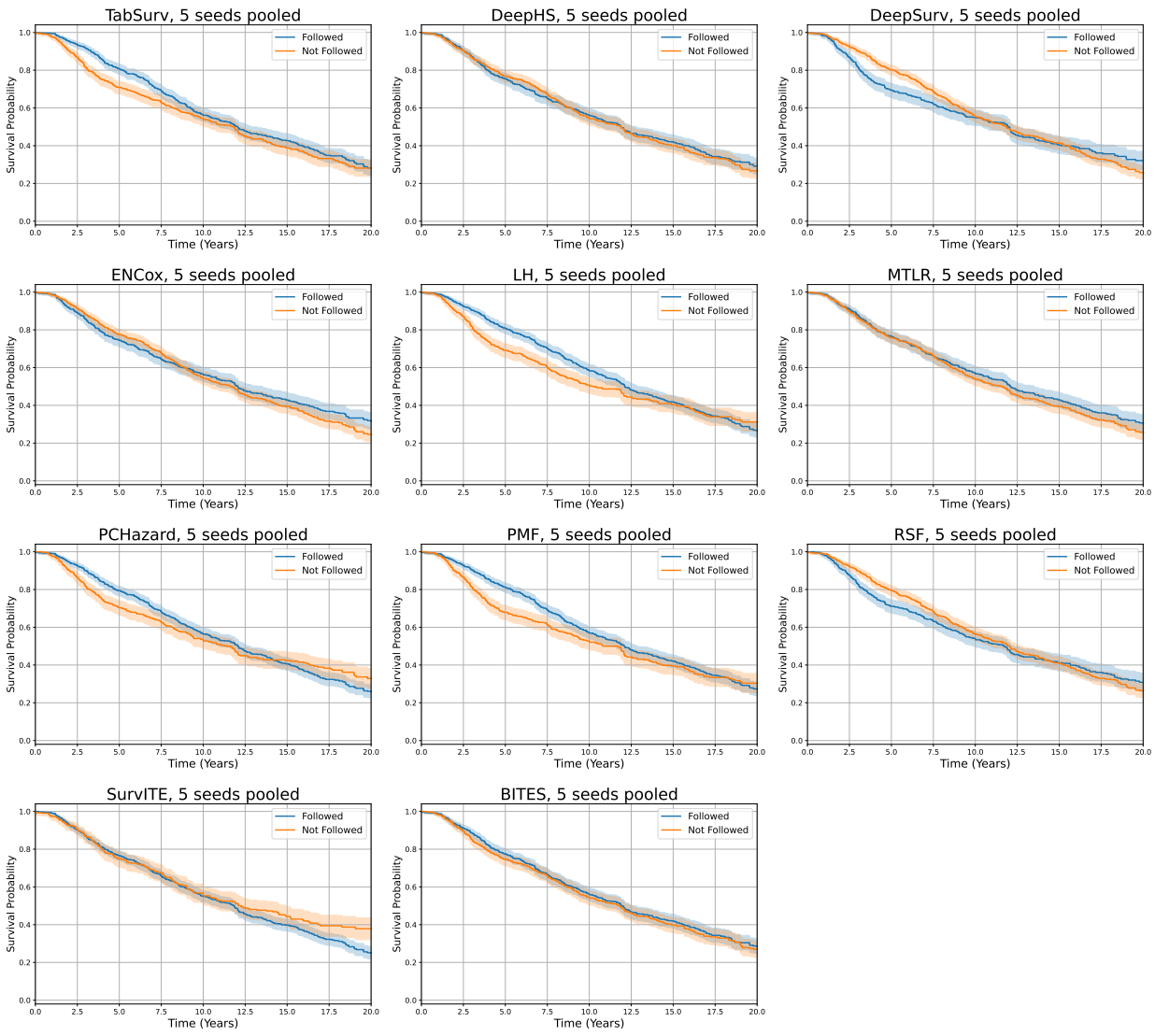


Figure S7: Kaplan–Meier survival curves for treatment recommendation using Nik-panel cancer driver genes. For TabSurv and each baseline model, patients are stratified into *Followed* and *Not Followed* groups based on whether the treatment actually received matches the model-recommended therapy.

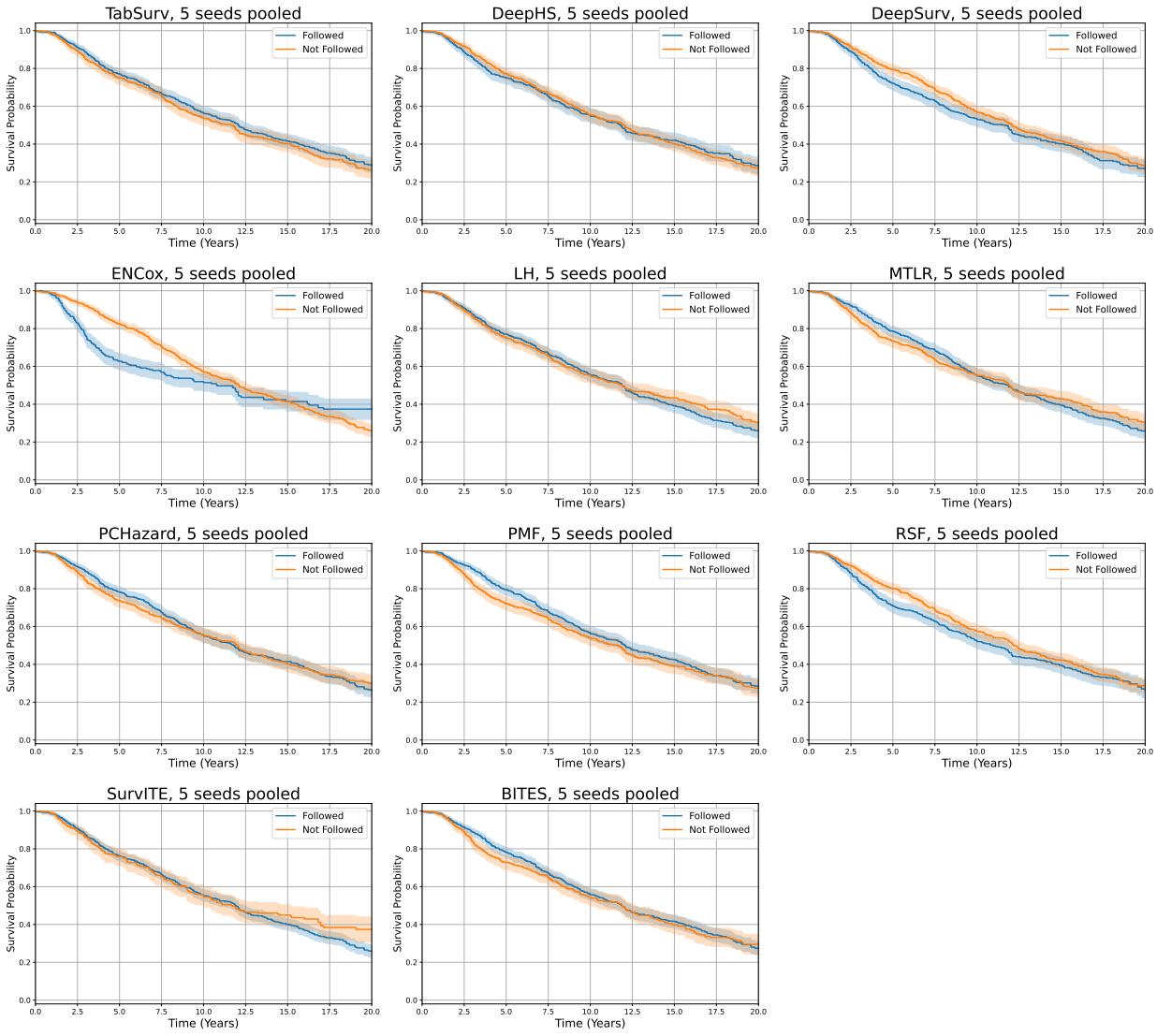


Figure S8: Kaplan–Meier survival curves for treatment recommendation using all genes. For TabSurv and each baseline model, patients are stratified into *Followed* and *Not Followed* groups based on whether the treatment actually received matches the model-recommended therapy.

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